

Laplace Transform Pairs — Full Derivation Cheat Sheet

(Companion to Table 3.4.1. All signals assume $x(t) = 0$ for $t < 0^-$ and $T \geq 0$.)

0. The Five Tools Used Throughout

Every entry in the table is built from the definition plus five reusable properties. Learn these once, and every row becomes a one- or two-line derivation.

Tool	Statement
Definition	$X(s) = \int_0^{\infty} x(t)e^{-st} dt$
Sifting	$\int f(t)\delta(t - t_0) dt = f(t_0)$
Linearity	$\mathcal{L}\{ax_1(t) + bx_2(t)\} = aX_1(s) + bX_2(s)$
Time-shift	$\mathcal{L}\{x(t - T)u(t - T)\} = e^{-Ts}X(s)$
Frequency-shift	$\mathcal{L}\{e^{-at}x(t)\} = X(s + a)$
Time-domain differentiation	$\mathcal{L}\{x'(t)\} = sX(s) - x(0)$
s-domain differentiation	$\mathcal{L}\{tx(t)\} = -\frac{dX(s)}{ds}$

Everything below is these seven facts, applied repeatedly.

$$1. \delta(t) \iff 1$$

Apply the sifting property directly inside the definition, with $f(t) = e^{-st}$:

$$X(s) = \int_{-\infty}^{\infty} \delta(t)e^{-st} dt = e^{-s(0)} = \boxed{1}$$

$$1a. \delta(t - T) \iff e^{-Ts}$$

Same sifting property, now evaluating e^{-st} at $t = T$:

$$X(s) = \int_{-\infty}^{\infty} \delta(t - T) e^{-st} dt = e^{-sT} = \boxed{e^{-Ts}}$$

(Alternatively: this is the time-shift property applied to Row 1, since $\delta(t - T)$ is $\delta(t)$ shifted by T .)

2. $u(t) \iff \frac{1}{s}$

Direct integration:

$$X(s) = \int_0^{\infty} (1) e^{-st} dt = \left[-\frac{e^{-st}}{s} \right]_0^{\infty} = 0 - \left(-\frac{1}{s} \right) = \boxed{\frac{1}{s}}$$

(valid for $\text{Re}(s) > 0$, so the upper limit vanishes)

2a. $u(t - T) \iff \frac{e^{-Ts}}{s}$

Apply the **time-shift property** to Row 2, with $X(s) = 1/s$:

$$\mathcal{L}\{u(t - T)\} = e^{-Ts} \cdot X(s) = \boxed{\frac{e^{-Ts}}{s}}$$

3. $e^{-at}u(t) \iff \frac{1}{s + a}$

Direct integration — combine the two exponentials before integrating:

$$X(s) = \int_0^{\infty} e^{-at} e^{-st} dt = \int_0^{\infty} e^{-(s+a)t} dt = \left[-\frac{e^{-(s+a)t}}{s + a} \right]_0^{\infty} = \boxed{\frac{1}{s + a}}$$

$$\mathbf{3a.} e^{-a(t-T)}u(t-T) \iff \frac{e^{-Ts}}{s+a}$$

Time-shift property applied to Row 3:

$$\mathcal{L}\{e^{-a(t-T)}u(t-T)\} = e^{-Ts} \cdot \frac{1}{s+a} = \boxed{\frac{e^{-Ts}}{s+a}}$$

$$\mathbf{4.} t u(t) \iff \frac{1}{s^2}$$

Apply **s-domain differentiation** to Row 2 ($X(s) = 1/s$):

$$\mathcal{L}\{t u(t)\} = -\frac{d}{ds} \left(\frac{1}{s} \right) = - \left(-\frac{1}{s^2} \right) = \boxed{\frac{1}{s^2}}$$

(Cross-check by parts: $\int_0^\infty t e^{-st} dt, u = t, dv = e^{-st} dt \Rightarrow \left[-\frac{t e^{-st}}{s} \right]_0^\infty + \frac{1}{s} \int_0^\infty e^{-st} dt = 0 + \frac{1}{s^2}$. Matches.)

$$\mathbf{4a.} (t-T)u(t-T) \iff \frac{e^{-Ts}}{s^2}$$

Time-shift property applied to Row 4:

$$\mathcal{L}\{(t-T)u(t-T)\} = e^{-Ts} \cdot \frac{1}{s^2} = \boxed{\frac{e^{-Ts}}{s^2}}$$

$$\mathbf{5.} t^2 u(t) \iff \frac{2}{s^3}$$

Apply **s-domain differentiation twice** to Row 2 (equivalently, once to Row 4):

$$\mathcal{L}\{t \cdot (t u(t))\} = -\frac{d}{ds} \left(\frac{1}{s^2} \right) = - \left(-\frac{2}{s^3} \right) = \boxed{\frac{2}{s^3}}$$

(General pattern: $\mathcal{L}\{t^n u(t)\} = \frac{n!}{s^{n+1}}$ — each differentiation brings down one more factorial and one more power of s in the denominator.)

$$6. t e^{-at} u(t) \iff \frac{1}{(s+a)^2}$$

Apply the **frequency-shift property** to Row 4. With $F(s) = 1/s^2$:

$$\mathcal{L}\{e^{-at} \cdot t u(t)\} = F(s+a) = \boxed{\frac{1}{(s+a)^2}}$$

$$7. t^2 e^{-at} u(t) \iff \frac{2}{(s+a)^3}$$

Frequency-shift property applied to Row 5. With $F(s) = 2/s^3$:

$$\mathcal{L}\{e^{-at} \cdot t^2 u(t)\} = F(s+a) = \boxed{\frac{2}{(s+a)^3}}$$

$$8. t^{n-1} e^{-at} u(t) \iff \frac{(n-1)!}{(s+a)^n}$$

Step 1 — establish the unshifted general pattern (extending Row 5's logic by induction). Differentiating $X(s) = 1/s$ in the s -domain k times brings out $k!$ and raises the power of s by k :

$$\mathcal{L}\{t^{n-1} u(t)\} = \frac{(n-1)!}{s^n}$$

Step 2 — apply the frequency-shift property, replacing $s \rightarrow s+a$:

$$\mathcal{L}\{e^{-at} t^{n-1} u(t)\} = \boxed{\frac{(n-1)!}{(s+a)^n}}$$

(Sanity checks: $n = 1 \Rightarrow \frac{0!}{(s+a)^1} = \frac{1}{s+a}$, matching Row 3. $n = 2 \Rightarrow \frac{1!}{(s+a)^2}$, matching Row 6. $n = 3 \Rightarrow \frac{2!}{(s+a)^3} = \frac{2}{(s+a)^3}$, matching Row 7.)

9. $\sin(\omega_0 t)u(t) \iff \frac{\omega_0}{s^2 + \omega_0^2}$

Step 1 — rewrite via Euler's formula: $\sin(\omega_0 t) = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}$

Step 2 — integrate term by term (each is Row 3 with a complex "a"):

$$X(s) = \frac{1}{2j} \left(\frac{1}{s - j\omega_0} - \frac{1}{s + j\omega_0} \right)$$

Step 3 — combine over a common denominator, using $(s - j\omega_0)(s + j\omega_0) = s^2 + \omega_0^2$:

$$X(s) = \frac{1}{2j} \cdot \frac{2j\omega_0}{s^2 + \omega_0^2} = \boxed{\frac{\omega_0}{s^2 + \omega_0^2}}$$

10. $\sin(\omega_0 t + \theta)u(t) \iff \frac{s \sin \theta + \omega_0 \cos \theta}{s^2 + \omega_0^2}$

Step 1 — expand with the angle-addition identity:

$$\sin(\omega_0 t + \theta) = \sin(\omega_0 t) \cos \theta + \cos(\omega_0 t) \sin \theta$$

Step 2 — apply **linearity**, using Row 9 for $\sin(\omega_0 t)$ and Row 11 (below) for $\cos(\omega_0 t)$:

$$X(s) = \cos \theta \cdot \frac{\omega_0}{s^2 + \omega_0^2} + \sin \theta \cdot \frac{s}{s^2 + \omega_0^2} = \boxed{\frac{s \sin \theta + \omega_0 \cos \theta}{s^2 + \omega_0^2}}$$

11. $\cos(\omega_0 t)u(t) \iff \frac{s}{s^2 + \omega_0^2}$

Step 1 — note that $\frac{d}{dt} \sin(\omega_0 t) = \omega_0 \cos(\omega_0 t)$.

Step 2 — apply **time-domain differentiation** to Row 9's signal $x(t) = \sin(\omega_0 t)u(t)$, noting $x(0) = 0$:

$$\mathcal{L}\{\omega_0 \cos(\omega_0 t)u(t)\} = sX(s) - x(0) = s \cdot \frac{\omega_0}{s^2 + \omega_0^2} - 0$$

Step 3 — divide both sides by ω_0 :

$$\boxed{\frac{s}{s^2 + \omega_0^2}}$$

$$\mathbf{12.} \cos(\omega_0 t + \theta)u(t) \iff \frac{s \cos \theta - \omega_0 \sin \theta}{s^2 + \omega_0^2}$$

Step 1 — angle-addition identity:

$$\cos(\omega_0 t + \theta) = \cos(\omega_0 t) \cos \theta - \sin(\omega_0 t) \sin \theta$$

Step 2 — linearity, using Row 11 and Row 9:

$$X(s) = \cos \theta \cdot \frac{s}{s^2 + \omega_0^2} - \sin \theta \cdot \frac{\omega_0}{s^2 + \omega_0^2} = \boxed{\frac{s \cos \theta - \omega_0 \sin \theta}{s^2 + \omega_0^2}}$$

$$\mathbf{13.} e^{-at} \sin(\omega_0 t)u(t) \iff \frac{\omega_0}{(s + a)^2 + \omega_0^2}$$

Frequency-shift property applied to Row 9. With $F(s) = \frac{\omega_0}{s^2 + \omega_0^2}$, substitute $s \rightarrow s + a$:

$$\boxed{\frac{\omega_0}{(s + a)^2 + \omega_0^2}}$$

$$14. e^{-at} \cos(\omega_0 t) u(t) \iff \frac{s + a}{(s + a)^2 + \omega_0^2}$$

Frequency-shift property applied to Row 11. With $F(s) = \frac{s}{s^2 + \omega_0^2}$, substitute $s \rightarrow s + a$ everywhere s appears, including the numerator:

$$\boxed{\frac{s + a}{(s + a)^2 + \omega_0^2}}$$

$$15. 2e^{-at} \cos(bt - \theta) u(t) \iff \frac{e^{j\theta}}{s + a + jb} + \frac{e^{-j\theta}}{s + a - jb}$$

Step 1 — write the cosine via Euler's formula:

$$\cos(bt - \theta) = \frac{e^{j(bt-\theta)} + e^{-j(bt-\theta)}}{2}$$

Step 2 — multiply by $2e^{-at}$ and distribute:

$$2e^{-at} \cos(bt - \theta) = e^{-at} e^{j(bt-\theta)} + e^{-at} e^{-j(bt-\theta)} = e^{-j\theta} e^{-(a-jb)t} + e^{j\theta} e^{-(a+jb)t}$$

Step 3 — transform each exponential term separately using Row 3 (with complex "a"):

$$\mathcal{L}\{e^{-(a-jb)t} u(t)\} = \frac{1}{s + a - jb}, \quad \mathcal{L}\{e^{-(a+jb)t} u(t)\} = \frac{1}{s + a + jb}$$

Step 4 — apply linearity and reassemble:

$$X(s) = e^{-j\theta} \cdot \frac{1}{s + a - jb} + e^{j\theta} \cdot \frac{1}{s + a + jb} = \boxed{\frac{e^{j\theta}}{s + a + jb} + \frac{e^{-j\theta}}{s + a - jb}}$$

$$15a. e^{-at} \cos(bt - \theta) u(t) \iff \frac{(s + a) \cos \theta + b \sin \theta}{(s + a)^2 + b^2}$$

Step 1 — this signal is exactly half of Row 15's signal, so its transform is half of Row 15's result:

$$X(s) = \frac{1}{2} \left(\frac{e^{j\theta}}{s + a + jb} + \frac{e^{-j\theta}}{s + a - jb} \right)$$

Step 2 — combine over the common denominator $(s + a + jb)(s + a - jb) = (s + a)^2 + b^2$:

$$X(s) = \frac{e^{j\theta}(s + a - jb) + e^{-j\theta}(s + a + jb)}{2[(s + a)^2 + b^2]}$$

Step 3 — expand the numerator and group real terms:

$$(s + a)(e^{j\theta} + e^{-j\theta}) + jb(e^{-j\theta} - e^{j\theta}) = (s + a) \cdot 2 \cos \theta + jb \cdot (-2j \sin \theta) = 2(s + a) \cos \theta + 2b \sin \theta$$

Step 4 — the factor of 2 cancels with the 2 in the denominator:

$$\boxed{\frac{(s + a) \cos \theta + b \sin \theta}{(s + a)^2 + b^2}}$$

$$\mathbf{16.} \frac{2t^{n-1}}{(n-1)!} e^{-at} \cos(bt - \theta) u(t) \iff \frac{e^{j\theta}}{(s + a + jb)^n} + \frac{e^{-j\theta}}{(s + a - jb)^n}$$

This is the general-order version of Row 15 — multiply the same exponential decomposition by $\frac{t^{n-1}}{(n-1)!}$ instead of leaving it plain.

Step 1 — start from Row 15, Step 2's decomposition and multiply both sides by $\frac{t^{n-1}}{(n-1)!}$:

$$\frac{2t^{n-1}}{(n-1)!} e^{-at} \cos(bt - \theta) = \frac{t^{n-1}}{(n-1)!} e^{-j\theta} e^{-(a-jb)t} + \frac{t^{n-1}}{(n-1)!} e^{j\theta} e^{-(a+jb)t}$$

Step 2 — transform each term using **Row 8** (with $n \rightarrow n$, complex "a", which states

$$\mathcal{L} \left\{ \frac{t^{n-1}}{(n-1)!} e^{-ct} u(t) \right\} = \frac{1}{(s + c)^n}:$$

$$\mathcal{L} \left\{ \frac{t^{n-1}}{(n-1)!} e^{-(a-jb)t} u(t) \right\} = \frac{1}{(s + a - jb)^n}, \quad \mathcal{L} \left\{ \frac{t^{n-1}}{(n-1)!} e^{-(a+jb)t} u(t) \right\} = \frac{1}{(s + a + jb)^n}$$

Step 3 — apply linearity with the coefficients $e^{-j\theta}$ and $e^{j\theta}$ from Step 1:

$$X(s) = \frac{e^{j\theta}}{(s + a + jb)^n} + \frac{e^{-j\theta}}{(s + a - jb)^n}$$

(Sanity check: setting $n = 1$ collapses $\frac{2t^0}{0!} = 2$, recovering Row 15 exactly.)

Summary: Which Tool Builds Which Row

Row	Built from	Tool applied
1, 1a	Definition	Sifting property
2	Definition	Direct integration
2a, 3a, 4a	Row 2, 3, 4	Time-shift
3	Definition	Direct integration
4, 5	Row 2	s-domain differentiation (1x, 2x)
6, 7	Row 4, 5	Frequency-shift
8	Row 2 pattern	s-domain differentiation ($n-1$ times) + frequency-shift
9	Definition	Euler's formula + direct integration
10, 12	Row 9, 11	Angle-addition identity + linearity
11	Row 9	Time-domain differentiation
13, 14	Row 9, 11	Frequency-shift
15	Definition	Euler's formula + direct integration (complex a)
15a	Row 15	Algebraic simplification (combine complex conjugate terms)
16	Row 8 pattern	Euler's formula + Row 8 with complex a

The big picture: everything in this table reduces to knowing *one* base transform ($e^{-at}u(t) \leftrightarrow \frac{1}{s+a}$, or $u(t) \leftrightarrow \frac{1}{s}$) and *four* structural moves — differentiate in s to bring down powers of t , shift in s to attach e^{-at} , shift in t to attach delay, and use Euler's formula to turn

sines/cosines into complex exponentials. Once those are second nature, no table entry needs to be memorized — it can always be rebuilt in under a minute.